

## Session :-2

### 1. What are the three main layers of software architecture? Write one line about each.

**Ans:** The three main layers of software architecture are:

- **Presentation Layer (UI):** Handles user interaction by displaying information and receiving input.
- **Business Logic Layer:** Processes data, applies business rules, and manages the application's core functionality.
- **Data Access Layer (Database Layer):** Stores, retrieves, and manages data from the database.

### 2. Define client and server in your own words with one real-life example.

**Ans: Client** - A client is a device or application that requests services or information from another computer over a network.

**Server** - A server is a computer or program that provides services, data, or resources to clients when requested.

Real-life example: When you open YouTube on your phone, your phone is the client because it requests videos, and YouTube's computers are the servers because they store and send the videos to your phone.

### 3. Write 2 differences each between desktop apps, web apps, and mobile apps.

**Ans:**

| Type                | Two Differences                                                                                                |
|---------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Desktop Apps</b> | 1. Installed directly on a computer.<br>2. Can often work without an internet connection.                      |
| <b>Web Apps</b>     | 1. Run in a web browser without installation.<br>2. Usually require an internet connection to access.          |
| <b>Mobile Apps</b>  | 1. Installed on smartphones or tablets.<br>2. Can use device features like the camera, GPS, and notifications. |

**4. Name any 3 databases and mention whether each is SQL or NoSQL.**

**Ans:**

| Database   | Type  |
|------------|-------|
| MySQL      | SQL   |
| PostgreSQL | SQL   |
| MongoDB    | NoSQL |